

# Enabling Natural Language Query of Oracle E-Business Suite Release 12.2, Leveraging OCI Generative AI

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Oracle Technical Brief

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## Purpose statement

This technical brief describes a prototype Oracle APEX application that enables natural language query of Oracle E-Business Suite data by leveraging Oracle Cloud Infrastructure (OCI) Generative Artificial Intelligence (AI).

This application utilizes advanced generative AI technology to provide content and insights. While we strive for accuracy and reliability in the output generated, know that AI-driven solutions are not always accurate and occasionally produce unexpected results.

Users should exercise their own judgment when interpreting the application's output and verify any generated content before making decisions or taking actions based on it. The use of this application is at your own discretion, and Oracle Corporation and its affiliates disclaim any liability for any damages caused by use of this software.

This application is a prototype provided for demonstration purposes. It is not intended as a reference implementation to be cloned or customized.

The application is currently available only in English and does not support user preferences such as time zone or date or number formats.

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## Prerequisites

- Complete all prerequisites listed under the Prerequisites section of the Oracle technical brief [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#).
- Apply all patches listed under section 4.2 *Apply Patches to the Oracle E-Business Suite Instance of Installing Oracle Enterprise Command Center Framework, Release 12.2 (Latest Version)* (Doc ID [2495053.1](#)).
- Verify ownership of a full-use Oracle database license<sup>1</sup>.
- Install and use [Oracle Autonomous Database](#) release 19c or later, either Shared or Dedicated.
- Subscribe to OCI Generative AI.
  - While [OCI's Select AI feature](#) supports multiple AI services including Cohere, OpenAI and Azure OpenAI, this solution has only been tested with OCI Generative AI.

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<sup>1</sup> The limited-use Oracle database license agreement included with an Oracle E-Business Suite license does not allow for the creation of an additional schema as outlined in this paper. Refer to the following license agreement: [https://www.oracle.com/contracts/docs/application\\_licensing\\_table\\_070571.pdf](https://www.oracle.com/contracts/docs/application_licensing_table_070571.pdf). To be fully compliant, a full-use Oracle Database Enterprise Edition license is required.

## Functional Overview

This technical brief demonstrates how to enable natural language to SQL query translation for Oracle E-Business Suite data.

The following screenshots show how a user can ask the system questions in English, and the system responds by displaying records that directly answer these questions.

The screenshot shows the 'Ask Oracle E-Business Suite (EBS)' interface. The prompt is: 'show the details of the workorders released in the last 2 weeks which are delayed.' The system has responded with a table of work orders.

| Job Name | Job Status | Job Open Quantity | Quantity Completed | Wo Delayed Days | Wo Delayed | Wo Start Delayed Days | Wo Start Delayed | Wo Completion Delayed Days | Wo Completion Delayed | Actual Release Date | Actual Completion Date | Sched Start Date | Sched Completion Date | Custo |
|----------|------------|-------------------|--------------------|-----------------|------------|-----------------------|------------------|----------------------------|-----------------------|---------------------|------------------------|------------------|-----------------------|-------|
| 446916   | Released   | 10                | 0                  | 2.65466         | Yes        | .20535                | Yes              | 2.65466                    | Yes                   | 22-Oct-2024         |                        | 22-Oct-2024      | 22-Oct-2024           |       |
| 446911   | Released   | 5                 | 0                  | 3.35086         | Yes        |                       | No               | 3.35086                    | Yes                   | 20-Oct-2024         |                        | 21-Oct-2024      | 21-Oct-2024           |       |
| 446909   | On Hold    | 2                 | 0                  | 4.65466         | Yes        | .90722                | Yes              | 4.65466                    | Yes                   | 20-Oct-2024         |                        | 20-Oct-2024      | 20-Oct-2024           |       |
| 446908   | Released   | 3                 | 0                  | 4.65466         | Yes        | .90693                | Yes              | 4.65466                    | Yes                   | 20-Oct-2024         |                        | 20-Oct-2024      | 20-Oct-2024           |       |
| 447907   | Released   | 5                 | 0                  | 1.65466         | Yes        | .15101                | Yes              | 1.65466                    | Yes                   | 23-Oct-2024         |                        | 23-Oct-2024      | 23-Oct-2024           |       |
| 446910   | Complete   | 0                 | 4                  | .92469          | Yes        | .9078                 | Yes              | .92469                     | Yes                   | 20-Oct-2024         | 20-Oct-2024            | 20-Oct-2024      | 20-Oct-2024           |       |
| 446914   | Released   | 5                 | 0                  | 2.1755          | Yes        | .63826                | Yes              | 2.1755                     | Yes                   | 21-Oct-2024         |                        | 21-Oct-2024      | 22-Oct-2024           |       |

Manufacturing Example. In the Ask Oracle E-Business Suite (EBS) page, a user entered the prompt *show the details of the work orders released in the last 2 weeks which are delayed*. The system responded with a list of work orders released in the last two weeks and are delayed, listed by Job Name.

The screenshot shows the 'Ask Oracle E-Business Suite (EBS)' interface. The prompt is: 'Show employees with more than 25 years of experience along with their qualifications'. The system has responded with a table of employees.

| Full Name                    | Years Of Experience | Qualification                                                  | Qualification Award Date |
|------------------------------|---------------------|----------------------------------------------------------------|--------------------------|
| Palmer, Mr. James John (Jim) | 25.3                | Bachelor Degree                                                | 01-JUL-1991              |
| Ryan, Mrs. Carol             | 25.3                | Bachelor of Human Resources - Bachelor Degree                  |                          |
| Lagrange, Mr. Marc           | 25.4                | Masters Degree                                                 | 30-JUN-1986              |
| Brown, Mr. David             | 25.6                | Certified Employee Benefits Professional                       |                          |
| Roberts, Mr. Jackson         | 25.6                | Bachelor of Human Resources - Bachelor Degree                  | 15-APR-1996              |
| Brown, Mr. David             | 25.6                | Bachelor Degree                                                |                          |
| Rundgren, Mr. Ted            | 25.8                | Theory on Molecular Biology - Published Works                  |                          |
| Van Wyk, Miss Brenda         | 26.4                | Human Resources Management - Doctorate Degree                  |                          |
| Carter, Mr. John             | 26.8                | Certified Payroll Certificate - Certified Payroll Professional |                          |
| Saunders, Ms. Jennifer Susan | 26.8                | Bachelor of Human Resources - Bachelor Degree                  | 15-MAR-1994              |

HCM Example 1. In the Ask Oracle E-Business Suite (EBS) page, a user entered the prompt *Show employees with more than 25 years of experience along with their qualifications*. The system responded with a list of employee names together with their years of experience and qualification.

The screenshot shows the 'Ask Oracle E-Business Suite (EBS)' interface. At the top, there's a navigation bar with 'ORACLE' logo and links for Home, Favorites, Settings, Worklist, and Logout. Below the navigation bar, the 'Prompt' field contains the text 'show their current position and supervisor information'. A search icon is next to the prompt. To the right of the prompt is a 'Domain' dropdown menu set to 'People'. Below the prompt is a 'New Conversation' button. The main area displays a table with the following data:

| Full Name                    | Years Of Experience | Qualification                                                  | Qualification Award Date | Current Position            | Supervisor Name    | Supervisor Id |
|------------------------------|---------------------|----------------------------------------------------------------|--------------------------|-----------------------------|--------------------|---------------|
| Palmer, Mr. James John (Jim) | 25.3                | Bachelor Degree                                                | 01-JUL-1991              | 100001.ADMINISTRATOR        | Mills, Pat         | 7884          |
| Ryan, Mrs. Carol             | 25.3                | Bachelor of Human Resources - Bachelor Degree                  |                          | Human Resources Advisor     | Roberts, Jackson   | 4210          |
| Lagrange, Mr. Marc           | 25.4                | Masters Degree                                                 | 30-JUN-1986              | Vendeur zone sud.1          | Lamaury, Thibault  | 1703          |
| Brown, Mr. David             | 25.6                | Certified Employee Benefits Professional                       |                          |                             |                    |               |
| Roberts, Mr. Jackson         | 25.6                | Bachelor of Human Resources - Bachelor Degree                  | 15-APR-1996              | Human Resources Manager     | Saunders, Jennifer | 4214          |
| Brown, Mr. David             | 25.6                | Bachelor Degree                                                |                          |                             |                    |               |
| Rundgren, Mr. Ted            | 25.8                | Theory on Molecular Biology - Published Works                  |                          | 240100.PROFESSOR OF BIOLOGY | Turner, Norman     | 4815          |
| Van Wyk, Miss Brenda         | 26.4                | Human Resources Management - Doctorate Degree                  |                          | 4000JHR Director            | Maroga, Jacob      | 11814         |
| Saunders, Ms. Jennifer Susan | 26.8                | Bachelor of Human Resources - Bachelor Degree                  | 15-MAR-1994              | Director of Human Resources | York, Alexander    | 4230          |
| Carter, Mr. John             | 26.8                | Certified Payroll Certificate - Certified Payroll Professional |                          | Payroll Officer             | Buckley, David     | 4348          |

On the right side, there's a 'Conversations' panel with 'Current' and 'Previous' tabs. The 'Current' tab shows two conversation items: 'Show employees with more than 25 years of experience along with their qualifications' and 'show their current position and supervisor information'.

HCM Example 2. In the Ask Oracle E-Business Suite (EBS) page, a user entered a follow-up question at the prompt *show their current position and supervisor information*. The system responded with the same employees but added their current positions and supervisor information.

## Clearing Conversation Context

Notice that most of the questions are conversational, enabling the next question to refer to the preceding question using anaphoric references. By default, OCI Generative AI assumes you are asking a follow-up question.

To clear the context and start a new line of questioning, click **New Conversation**.

The screenshot shows the 'Ask Oracle E-Business Suite (EBS)' interface. The 'Prompt' field contains the text 'Type a question'. A search icon is next to the prompt. Below the prompt is a 'New Conversation' button, which is highlighted with a red box. To the right of the prompt is a 'Domain' dropdown menu set to 'People'.

## Selecting a Domain

If you know the business domain of your question, you can select it from the **Domain** drop-down list. This provides a hint to the generative AI and may produce a more accurate answer to your question.

The screenshot shows the 'Ask Oracle E-Business Suite (EBS)' interface. The 'Prompt' field contains the text 'Type a question'. A search icon is next to the prompt. Below the prompt is a 'New Conversation' button. To the right of the prompt is a 'Domain' dropdown menu set to 'Discrete Manufacturing', which is highlighted with a red box.

## Viewing the Result SQL

For diagnostic purposes, it is useful to view the SQL statement generated by the OCI Generative AI service. The administrator may enable a **Show SQL** button for a particular user. This button is disabled by default but can be turned on by a user-level profile option **FND: NLQ Show SQL**.

**Ask Oracle E-Business Suite (EBS)**

Prompt  
show me all work orders

Domain  
Discrete Manufacturing

New Conversation ?

show me all

**SQL Query**

```
WITH dt as (SELECT JOB_NAME "Job Name", JOB_STATUS "Job Status", JOB_DESCRIPTION "Job Description", SCHED_START_DATE "Scheduled Start Date", SCHED_COMPLETION_DATE "Scheduled Completion Date", ACTUAL_RELEASE_DATE "Actual Release Date", ACTUAL_COMPLETION_DATE "Actual Completion Date", QUANTITY_COMPLETED "Quantity Completed", WO_DELAYED "Workorder Delayed", WO_START_DELAYED "Workorder Start Delayed", WO_COMPLETION_DELAYED "Workorder Completion Delayed", CUSTOMER "Customer", SALES_ORDER "Sales Order", SO_LINE_NUMBER "Sales Order Line Number", SHIP_DATE "Ship Date", ACTUAL_SHIP_DATE "Actual Ship Date" FROM XX_NLQ.WIP_NLQ_DWO_V) SELECT /*+ FIRST_ROWS(100)* / DISTINCT dt.* from dt
```

Show SQL ↗

| Job Name | Job Status | Job Description | Scheduled Start Date | Scheduled Completion Date | Actual Release Date | Actual Completion Date | Quantity Completed | Workorder Delayed | Workorder Start Delayed | Workorder Completion Delayed |
|----------|------------|-----------------|----------------------|---------------------------|---------------------|------------------------|--------------------|-------------------|-------------------------|------------------------------|
| TEST_KC  | Released   |                 |                      |                           |                     |                        | 0                  | No                | No                      | No                           |
| 395909   | On Hold    |                 | 2024                 | 2024                      |                     |                        |                    | No                | No                      | No                           |
| 395911   | Released   |                 | 28-Oct-2024          | 31-Oct-2024               | 28-Oct-2024         |                        | 0                  | Yes               | Yes                     | No                           |
| 395910   | On Hold    |                 | 28-Oct-2024          | 31-Oct-2024               | 28-Oct-2024         |                        | 0                  | Yes               | Yes                     | No                           |
| 395908   | Complete   |                 | 28-Oct-2024          | 29-Oct-2024               | 28-Oct-2024         | 28-Oct-2024            | 5                  | No                | Yes                     | No                           |

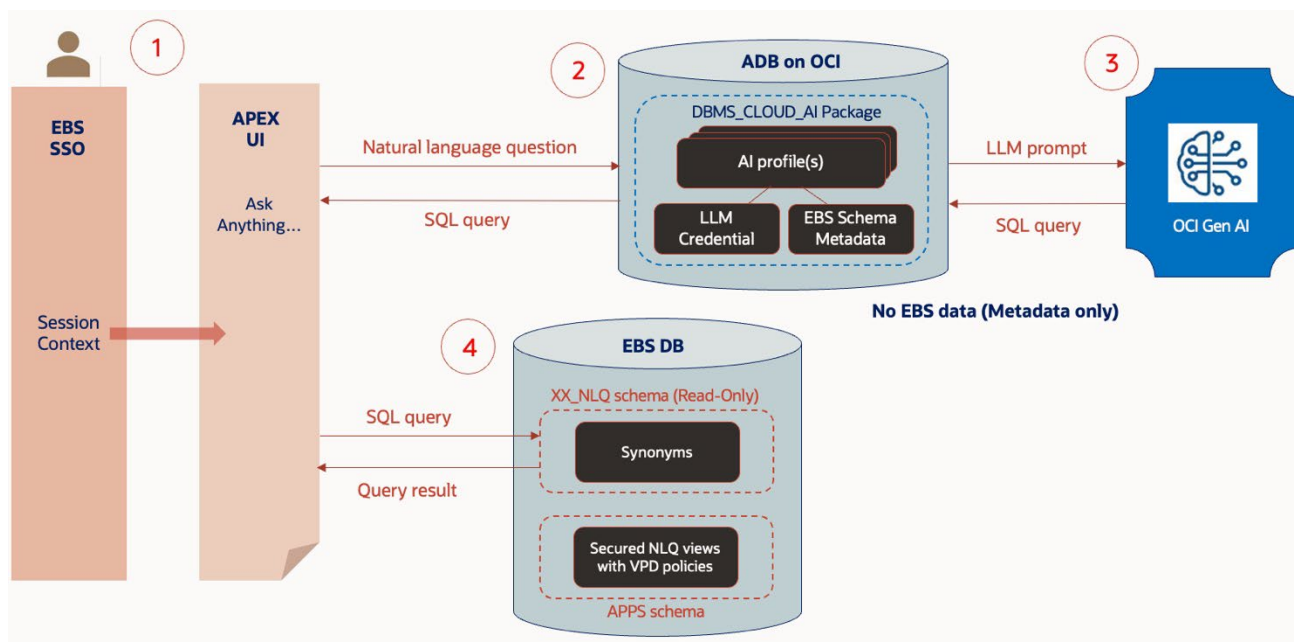
To enable this button for a user:

1. Log in to E-Business Suite as SYSADMIN.
2. Navigate to **Functional Administrator**.
3. Click **Core Services** then click **Profiles**.
4. In the **Code** field, enter **FND\_NLQ\_SHOW\_SQL** then click **Go**.
5. Click **Update Value**.
6. Click the **User** tab.
7. Click **Add**.
8. In the **User** field, enter the user for whom to enable this feature.
9. In the **Value** field, enter **Yes**.
10. Click **Update**.



## Technical Overview

The following architecture diagram shows the interaction between the technical components used.



This Architecture Diagram shows the interaction between the APEX Frontend, ADB on OCI, the LLM Service, and EBS.

### 1. APEX Frontend

This implementation uses Oracle APEX to develop the user interface and integrate with Oracle E-Business Suite (EBS) Single Sign-On (SSO) for user authentication. The APEX app is embedded within OA Framework and ensures that the EBS security setup is honored. In addition to SSO, standard EBS FND\_USER authentication is also supported.

The APEX UI sends the natural language question to the Oracle Autonomous Database (ADB) using its Select AI API (DBMS\_CLOUD\_API) package. The ADB returns a SQL query that addresses the question.

### 2. Oracle Autonomous Database

Upon receipt of a natural language question, the ADB's Select AI feature generates an appropriate prompt and sends it to the Oracle Generative AI service.

The ADB is configured with several AI profiles. You can think of an AI profile as a domain expert that answers questions for a particular business domain. Technically, an AI profile is an ADB object that contains a subset of EBS metadata (essentially, view names) that can best solve the questions in its domain.

For example, the sample software (Refer to Downloading the Sample Software) contains the following AI profiles to answer questions for a corresponding business domain:

| AI PROFILE NAME        | BUSINESS DOMAIN        |
|------------------------|------------------------|
| EBS_NLQ_DMFCST_PROFILE | Discrete Costing       |
| EBS_NLQ_DMF_PROFILE    | Discrete Manufacturing |
| EBS_NLQ_AR_PROFILE     | Dispute                |
| EBS_NLQ_INV_PROFILE    | Logistics              |



|                     |                       |
|---------------------|-----------------------|
| EBS_NLQ_EAM_PROFILE | Maintenance           |
| EBS_NLQ_ONT_PROFILE | Order Management      |
| EBS_NLQ_AP_PROFILE  | Payment               |
| EBS_NLQ_HCM_PROFILE | People                |
| EBS_NLQ_GME_PROFILE | Process Manufacturing |
| EBS_NLQ_PJM_PROFILE | Project Manufacturing |

Given a natural language question, the sample code's logic chooses the most relevant AI profile that has the highest probability of answering the question correctly. It then instructs the ADB to use this profile to compose a prompt. The SQL statement returned by the generative AI service's Large Language Model (LLM) contains views and columns constrained by this AI profile. The AI profile concept provides a way to constrain the number of database objects that the LLM must analyze, resulting in better performance and accuracy.

Notice that the ADB does not contain any data, only schema metadata (such as view definitions), so your data is never exposed to any third-party systems.

### 3. Generative AI Large Language Model

Ultimately, the OCI Generative AI LLM is responsible for constructing a valid SQL query.

### 4. E-Business Suite Database

The code provided checks the syntax of the SQL response received from the LLM by way of the Autonomous Database Select AI API return value. If the SQL response is valid, APEX forwards the response to the EBS database for execution. A table displays the records returned by the SQL response. The SQL executes in a new custom schema with the minimally required READ privileges to provide maximum security. Virtual Private Databases (VPDs) guarantee that the end user has the authority to view the returned records. See the [Security Overview](#) for a more detailed explanation.

## Security Overview

The following software security best practices have been considered during the integration of Oracle E-Business Suite (EBS) with an OCI Generative AI service provider:

- Separation of Duties
- Principle of Least Privilege
- Single Sign-On Authentication
- Role-Based Access Control
- Record-Level Data Security
- Privacy by Design
- Audit Logging

### Separation of Duties

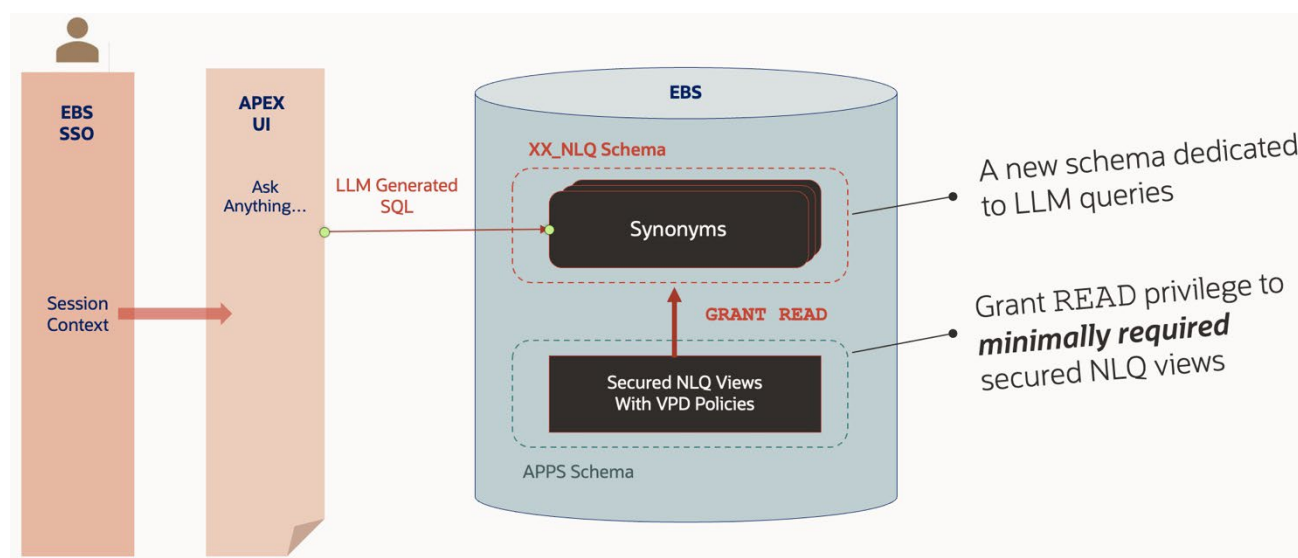
Separation of Duties addresses the concern of exposing your data to a third-party AI service provider by separating tasks into two distinct duties:

1. Converting a natural language question into a valid SQL query.
2. Executing the SQL query to retrieve the results.

As shown in the previous [Architecture Diagram](#), the above two duties are achieved by two distinct code paths. The top half of the diagram is responsible for constructing the SQL. In this pathway, the AI service provider is contacted by the Oracle Autonomous Database, but there is no customer data exchanged, only EBS metadata (view definitions). When the returned SQL query is executed as illustrated in the bottom half of the diagram, the AI service provider is no longer involved. This facilitates the separation of duties principle.

### Principle of Least Privilege

To prevent the SQL from querying all data and from using SQL commands such as INSERT, UPDATE, DELETE, or DROP TABLE, a new schema (named XX\_NLQ) is created in your database. This new schema has minimal privileges, as illustrated in the following diagram:



XX\_NLQ Schema: The new XX\_NLQ Schema in EBS is defined with minimal privileges.

### Single Sign-On Authentication

This implementation uses Oracle E-Business Suite's authentication setup to authenticate the user. This is achieved by a new OA Framework enhancement. For more information about this enhancement, refer to this Oracle technical brief: [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#)

### **Role-Based Access Control**

This implementation uses Oracle E-Business Suite's comprehensive Role-Based Access Control (RBAC) security feature to grant a user access to an AI profile. You can grant access either explicitly or indirectly, by granting AI profile access to a Responsibility. See [Configuring Security](#).

For more information about the general EBS RBAC feature, see Oracle Support Document [1537100.1](#), *Function Security and Role-Based Access Control (RBAC) in Oracle E-Business Suite*.

### **Record-Level Data Security**

This implementation provides fine-grain record-level security. Each data source is a view with an accompanying PL/SQL security handler. By creating a Virtual Private Database (VPD) policy on the data source and then performing a SELECT command, the database engine calls the PL/SQL security handler, which returns a WHERE clause that filters out all unauthorized records from the view. Because VPD is an underlying database technology, there is no complex, manually-coded security logic to introduce security vulnerabilities.

### **Privacy by Design**

Oracle Development curates the data sources and takes care during the design phase to ensure that columns containing critical data, such as HIPAA and PCI information, are excluded from the views. This eliminates the risk of accidentally exposing such data. Even if the LLM constructs a query to select from those columns, the query fails since no such columns exist in the curated views. The secured views only select the minimally required columns from the base views. Since the query is performed in the XX\_NLQ Schema, there is no possibility of extracting critical data from the database.

### **Audit Logging**

The implementation logs every question and the results (SQLs) in a logging table. This logging table serves the following two purposes:

- The data can be used for debugging and improving accuracy.
- A DBA can review the data to determine if there are any malicious attempts or misuse of the solution.

The table is available in the Autonomous Database schema XX\_NLQ\_UTIL. The table name is FND\_NLQ\_SESSION\_HISTORY.

## Subscribing to OCI Generative AI

This implementation uses Oracle Cloud Infrastructure (OCI) Generative AI to perform natural language to SQL translation.

### 1. Subscribe to OCI Generative AI.

To use OCI Generative AI, subscribe to an OCI region that offers OCI Generative AI. Refer to [Subscribing to an Infrastructure Region](#) for more information and instructions.

Not all OCI regions support Generative AI. Refer to [Regions with Generative AI](#) for more information.

### 2. Note the OCI Generative AI API Endpoint Information.

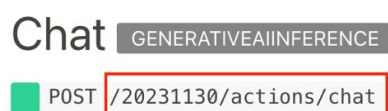
During the installation, you will be prompted to enter your OCI Generative AI API information. Note the following information:

1. Refer to the [Generative AI Service Inference API](#) page and note the API Endpoint of your subscribed region. This will be referred to as the `ocigenai_endpoint` in a subsequent step. Currently, it can be one of the following URLs:

#### API Endpoints:

- <https://inference.generativeai.eu-frankfurt-1.oci.oraclecloud.com>
- <https://inference.generativeai.sa-saopaulo-1.oci.oraclecloud.com>
- <https://inference.generativeai.uk-london-1.oci.oraclecloud.com>
- <https://inference.generativeai.us-chicago-1.oci.oraclecloud.com>

From the [OCI Generative AI Chat API](#) page, note the POST API name. This will be referred to as `ocigenai_chat_api` in a subsequent step. Currently, it is shown as `/20231130/actions/chat`:



**Important:** The leading forward slash is part of the API signature. Do not omit the leading forward slash.

### 3. (Highly Recommended) Create a Service Gateway.

The OCI Generative AI REST endpoint resides in the public internet. If you do not want any network traffic exposed to the public internet, create a Service Gateway in your OCI Virtual Cloud Network (VCN). See [Access to Oracle Services: Service Gateway](#) for more information and instructions.

Additionally, when you create the Oracle Autonomous Database instance, you must create it in the same VCN and region as the OCI Generative AI. This ensures that the traffic from the VCN to the OCI Generative AI service travels over the Oracle network instead of the internet.

## Downloading the Sample Software

Download the [EBS NLQ Code <date stamp>.zip](#) file containing the sample software from My Oracle Support document [3059877.1](#), *Enabling Natural Language Query of Oracle E-Business Suite of Release 12.2, Leveraging Oracle Generative AI*.

## Configuring the Oracle Autonomous Database

### 1. Create an Autonomous Database instance.

Use the OCI dashboard to create an Autonomous Database instance with the following characteristics:

- A database version of 19c or later.
- Serverless or Dedicated.
- (Recommended) The Transaction Processing variant selected.
- (Recommended) The Private Endpoint Access Only access type selected.
- (Recommended) The Autonomous Database instance in the same region as the OCI Generative AI service.

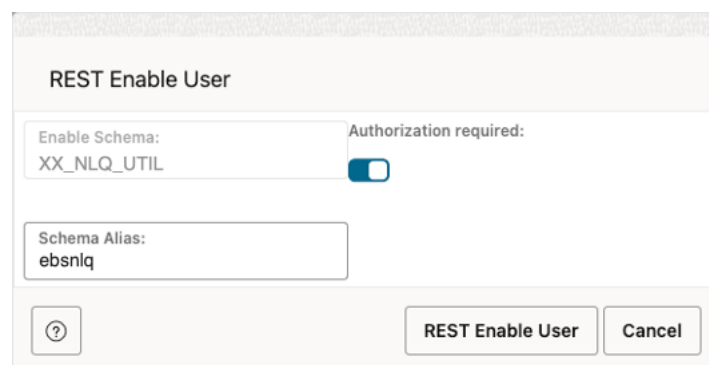
### 2. Create an XX\_NLQ\_UTIL schema.

A new XX\_NLQ\_UTIL schema is required to install *Select AI* metadata, some query code, and common tables, such as history and logging. Log in as the Autonomous Database ADMIN user and run the following commands:

```
CREATE USER xx_nlq_util IDENTIFIED BY <your password> QUOTA 10G ON DATA;  
  
GRANT CREATE SESSION TO xx_nlq_util;  
  
GRANT CREATE TABLE TO xx_nlq_util;  
  
GRANT CREATE VIEW TO xx_nlq_util;  
  
GRANT CREATE ANY INDEX TO xx_nlq_util;  
  
GRANT CREATE SEQUENCE TO xx_nlq_util;  
  
GRANT CREATE PROCEDURE TO xx_nlq_util;  
  
GRANT EXECUTE ON dbms_cloud_ai TO xx_nlq_util;  
  
GRANT EXECUTE ON dbms_cloud TO xx_nlq_util;
```

### 3. Enable REST for XX\_NLQ\_UTIL.

Log in to the Autonomous Database admin console. From the **Administration** menu, select **Database Users**. For the XX\_NLQ\_UTIL user, choose the drop-down icon. Select **Enable REST**. In the **Schema Alias** field, enter **ebsnlq**, and turn on **Authorization required**:



The screenshot shows a modal dialog titled "REST Enable User". It contains three input fields: "Enable Schema:" with the value "XX\_NLQ\_UTIL", "Authorization required:" with a checked toggle switch, and "Schema Alias:" with the value "ebsnlq". At the bottom, there is a help icon (question mark), a "REST Enable User" button, and a "Cancel" button.

### 4. Configure OCI Generative AI.

To set up OCI Generative AI, follow these steps:

**Tip:** For more information about these steps, refer to [Perform Prerequisites to Use Resource Principal with Autonomous Database](#) and [Integrating OCI Generative AI with Select AI and APEX to query data using natural language](#).

1. Create a dynamic group.
  - a. Log in to your OCI admin console.
  - b. From the navigation menu, select **Identity & Security**. Under **Identity**, select **Domains**.
  - c. Click the name of the identity domain that you want to work in.  
**Tip:** You might need to change the compartment to find the domain that you want.
  - d. Click **Dynamic Groups**.
  - e. Click **Create dynamic group**.
  - f. Create a rule that includes the Autonomous Database resource as a member of this group.  
For more information about this step, refer to [Creating a Dynamic Group, Use Resource Principal to Access Oracle Cloud Infrastructure Resources](#) and [Writing Matching Rules to Define Dynamic Groups](#).

For example, in the **Matching rules** page, add the following rule:

```
resource.id = '<your-autonomous-database-ocid>'
```

2. Set the required Identity and Access Management (IAM) policy.
  - a. From the navigation menu, select **Identity & Security**. Under **Identity**, select **Policies**.
  - b. Click **Create Policy**.
  - c. Enter a name and a description.
  - d. In the **Policy Builder**, enter the following policy:  
allow dynamic-group <your-group-name> to manage generative-ai-family in compartment <your-compartment-name>
  - e. Note the OCID of the compartment-name. Use the OCID in a subsequent step when you install the script `adb/scripts/nlq_util_seeddata.sql`.

3. Connect as an ADMIN user to the autonomous database and enable the OCI resource principal authentication:

```
EXEC DBMS_CLOUD_ADMIN.ENABLE_RESOURCE_PRINCIPAL();  
EXEC DBMS_CLOUD_ADMIN.ENABLE_RESOURCE_PRINCIPAL(username => 'XX_NLQ_UTIL');
```

**Note:** Due to the caching of the resource principal token, it may take up to two hours for your changes to take effect.

## 5. (Conditional) Update the OCI Generative AI Model.

One of the scripts sets up meta.llama-3.1-70b-instruct as the LLM model. Since an LLM model may be deprecated and replaced by a more advanced version, consult the *Chat Models* section in [Pretrained Foundational Models in Generative AI](#). If the meta.llama-3.1-70b-instruct model is no longer available, note the current version and update the `l_llm_model` variable in the script to reflect the latest value.

The script name is `adb/scripts/nlq_util_ai_profiles.sql`.

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In the following code fragment of the file, change the `l_llm_model` variable to the current value. Create a backup before changing the file.

```
l_llm_provider  VARCHAR2(30) := 'oci';  
l_llm_model    VARCHAR2(30) := 'meta.llama-3.1-70b-instruct';
```

## 6. Install software and metadata.

1. Connect to the `XX_NLQ_UTIL` schema on the autonomous database.
2. Run each of the following scripts included in your download to install the software and metadata.

`adb/scripts/nlq_util_datamodel.sql`

`adb/scripts/nlq_util_seeddata.sql`

**Tip:** When loading this script to the database, enter these values when prompted:

- For `adb_compartment_ocid`, enter the compartment OCID noted in an [earlier step](#).
- For `ocigenai_endpoint`, enter the OCI Generative AI Endpoint URL noted in an [earlier step](#).
- For `ocigenai_chat_api`, enter the OCI Generative AI Chat API noted in an [earlier step](#).

**Important:** The leading forward slash is part of the API signature. Do not omit the leading forward slash.

`adb/scripts/nlq_util_core_s.sql`

`adb/scripts/nlq_util_core_b.sql`

`adb/scripts/nlq_util_metadata_ap.sql`

`adb/scripts/nlq_util_metadata_ar.sql`

`adb/scripts/nlq_util_metadata_dmf.sql`

`adb/scripts/nlq_util_metadata_eam.sql`

`adb/scripts/nlq_util_metadata_icx.sql`

`adb/scripts/nlq_util_metadata_inv.sql`

`adb/scripts/nlq_util_metadata_pjm.sql`

`adb/scripts/nlq_util_metadata_hcm.sql`

`adb/scripts/nlq_util_metadata_opm.sql`

`adb/scripts/nlq_util_metadata_ont.sql`

`adb/scripts/nlq_util_metadata_cst.sql`

`adb/scripts/nlq_util_ai_profiles.sql`

`adb/scripts/nlq_util_web_services.sql`

## 7. (Optional) Revoke grants.

After a successful installation, the following grants are no longer required. You can revoke them if desired to provide better security.

```
REVOKE CREATE TABLE FROM xx_nlq_util;
```

```
REVOKE CREATE VIEW FROM xx_nlq_util;
```

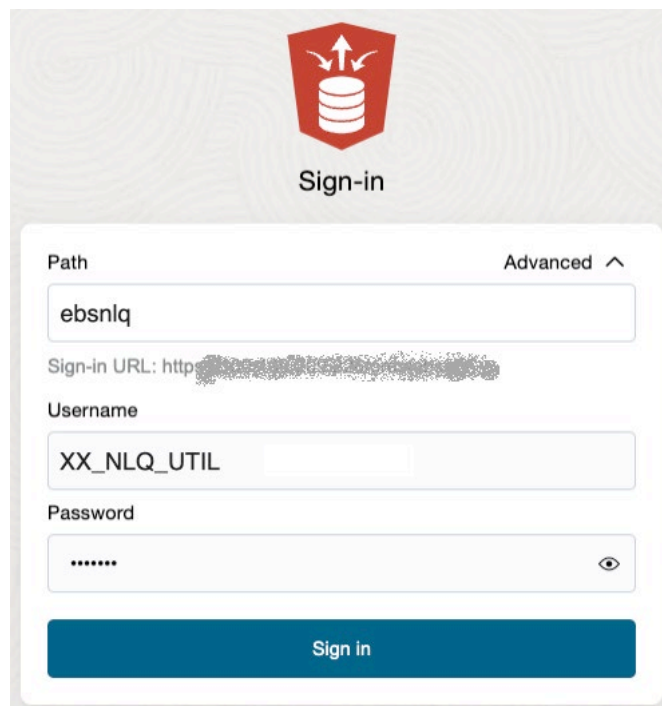
```
REVOKE CREATE ANY INDEX FROM xx_nlq_util;
```

```
REVOKE CREATE SEQUENCE FROM xx_nlq_util;
```

```
REVOKE CREATE PROCEDURE FROM xx_nlq_util;
```

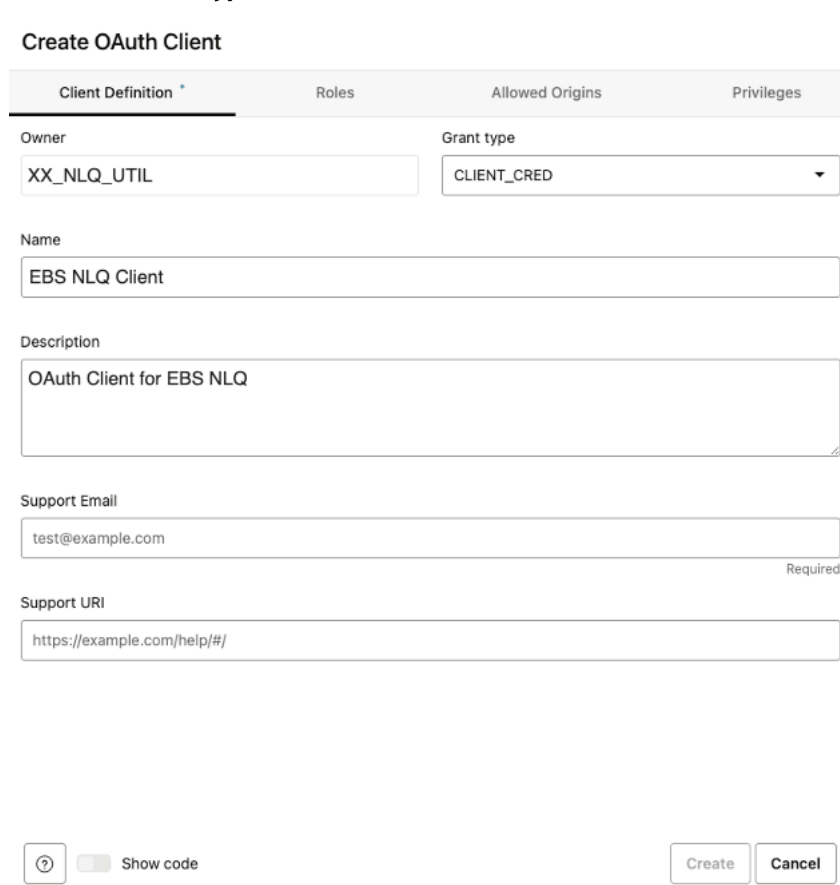
## 8. Create OAuth Client.

1. Log in to the Autonomous Database console as the `XX_NLQ_UTIL` user.
2. Expand the **Advanced** region. In the **Path** field, enter `ebsnlq`.



The image shows a 'Sign-in' dialog box with a red shield icon containing a database cylinder and an upward arrow. The dialog has a 'Path' field with the value 'ebsnlq' and an 'Advanced' toggle. Below the path is a 'Sign-in URL' field with a partially visible URL. There are 'Username' and 'Password' fields. The username is 'XX\_NLQ\_UTIL' and the password is masked with dots. A 'Sign in' button is at the bottom.

3. Navigate to REST, and then select CLIENTS.
4. Click **Create OAuth Clients**.
5. Select `CLIENT_CRED` for the **Grant type**, and then enter the **Name** and other details for the client.



The image shows the 'Create OAuth Client' form. It has a tabbed interface with 'Client Definition' selected. The form includes fields for 'Owner' (XX\_NLQ\_UTIL), 'Grant type' (CLIENT\_CRED), 'Name' (EBS NLQ Client), 'Description' (OAuth Client for EBS NLQ), 'Support Email' (test@example.com), and 'Support URI' (https://example.com/help/#/). There is a 'Required' label next to the Support Email field. At the bottom, there is a 'Show code' toggle and 'Create' and 'Cancel' buttons.

6. Under **Roles**, select *EBS NLQ Services Role*.
7. Under **Privileges**, select *ebsnlq.actions.generateSql.privilege* and *ebsnlq.actions.feedback.privilege*.
8. Click **Create**.
9. Note the **Client ID** and the **Client Secret**. You will use these when configuring the Oracle E-Business Suite AI APEX application.

For more information about this step, refer to [Creating an OAuth Client](#).

## Configuring Oracle E-Business Suite

**Important:** When running database installation scripts, note the schema connection details. Some scripts connect to SYSDBA, and some scripts connect to APPS. Run each script in their corresponding schema for a successful installation.

### 1. Apply required prerequisite patches.

Refer to the *Prerequisites* section of the Oracle technical brief [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#) and apply all the required EBS patches.

### 2. (Conditional) Install Oracle Enterprise Command Center database objects.

This example EBS Generative AI solution makes use of some Oracle Enterprise Command Center (ECC) database objects.

If you have implemented ECC Release 12.2, version 12 or later, skip to the next step.

If you are using ECC version 11 or earlier, you must update to ECC version 12 or later. Oracle recommends that you update to the latest available version. See *Installing Oracle Enterprise Command Center Framework, Release 12.2 (Latest Version)* (Doc ID [2495053.1](#)), Appendix C.

If you are currently not an ECC user, you must install the ECC database objects. You do not need to implement the full ECC functionality. Follow only the steps in section 4.2 *Apply Patches to the Oracle E-Business Suite Instance* of *Installing Oracle Enterprise Command Center Framework, Release 12.2 (Latest Version)* (Doc ID [2495053.1](#)).

### 3. Create an XX\_NLQ schema.

Connect to your E-Business Suite database as the SYSDBA user to create a new XX\_NLQ schema with the following minimum grants. If you are using Oracle Multitenancy architecture, connect to the E-Business Suite Pluggable Database.

```
CREATE USER xx_nlq IDENTIFIED BY <your password>;  
GRANT CREATE SESSION TO xx_nlq;
```

### 4. Register the XX\_NLQ schema with Oracle E-Business Suite.

UNIX:

```
$ . <RUN_BASE>/EBSapps/appl/APPS<CONTEXT_NAME>.env  
$ sqlplus apps/<APPS password> @$AD_TOP/patch/115/sql/ADAPEXSCHREG.sql XX_NLQ
```

Windows:

```
C:\> <RUN_BASE>\EBSapps\appl\envshell<CONTEXT_NAME>.cmd  
C:\> sqlplus apps/<APPS password> @%AD_TOP%\patch\115\sql\ADAPEXSCHREG.sql XX_NLQ
```

Enter the EBS\_SYSTEM schema password, APPS schema username, and APPS schema password when prompted.

### 5. Create database artifacts.

Connect to your E-Business Suite database as the APPS user and run the following scripts:

```
ebs/scripts/nlq_views.sql  
ebs/scripts/nlq_utils_s.sql  
ebs/scripts/nlq_utils_b.sql  
ebs/scripts/nlq_datasec_functions.sql  
ebs/scripts/nlq_vpd_policies.sql  
ebs/scripts/nlq_permissions.sql
```

ebs/scripts/nlq\_ebs\_profile.sql

## Configuring Oracle APEX

### 1. Deploy an Oracle APEX development platform.

Refer to the *Installing Oracle APEX* section of the Oracle technical brief [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#) to install and deploy Oracle APEX and Oracle REST Data Services.

### 2. Create an APEX workspace.

Refer to the Oracle technical brief [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#), section *Defining the Oracle APEX Workspace and Schema*, subsection *Creating the EBS\_APEX Workspace*, to create a new APEX Workspace named **EBS\_NLQ\_APEX**.

Use the following workspace name and schema name, which are different from the example given in the above technical brief.

For Workspace Name, enter **EBS\_NLQ\_APEX**.

For Reuse Existing Schema, choose **Yes**.

For Schema Name, enter **XX\_NLQ**.

**Note:** You do not need to follow the subsections *Creating the Custom Schema XX\_APEX* and *Registering XX\_APEX with Oracle E-Business Suite* since you are reusing the custom schema XX\_NLQ created and registered in the previous [step](#).

The screenshot displays the 'Create Workspace' wizard in the Oracle APEX interface. The top navigation bar includes 'APEX', 'Manage Requests', 'Manage Instance', 'Manage Workspaces', and 'Monitor Activity'. The breadcrumb trail shows 'Manage Workspaces > Create Workspace'. The wizard title is 'Create Workspace'. A progress bar at the top indicates four steps: 'Workspace Information' (completed), 'Administrator Information' (completed), 'Database Schema Information' (completed), and 'Confirm Request' (current step). The 'Workspace Information' section contains the following fields: 'Workspace name' (EBS\_NLQ\_APEX), 'Workspace ID' (System Assigned), and 'Description' (empty). The 'Administrator Information' section contains 'User name' (admin) and 'E-Mail' (empty). The 'Database Schema Information' section contains 'Reuse Existing Schema' (Yes) and 'Schema Name' (XX\_NLQ). At the bottom, there are 'Cancel' and 'Create Workspace' buttons.

After creating the APEX Workspace, you do not need to follow the other steps in the Oracle technical brief [Extending Oracle E-Business Suite Release 12.2 Using Oracle APEX](#).

### 3. Install the Oracle E-Business Suite Natural Language Query APEX application.

The following steps explain how to import the Oracle E-Business Suite Natural Language Query APEX application script file.

#### 21 Enabling Natural Language Query of Oracle E-Business Suite Release 12.2, Leveraging OCI Generative AI

1. Log in to APEX (https://<EBS hostname>:<port>/ords/apex):
  - a. In the Workspace field, enter **EBS\_NLQ\_APEX**.
  - b. Use your administrator username and password.
2. From the App Builder, click **Import**.
3. Select or drag and drop the downloaded application script file:
   
ebs/apex/apex\_ebs\_nlq\_extension.sql
4. Click **Next**, then click **Install Application**.

#### 4. Configure Network Access Control to the Autonomous Database.

Connect to your E-Business Suite database as the SYSDBA user and enable Network Access Control List for the APEX application.

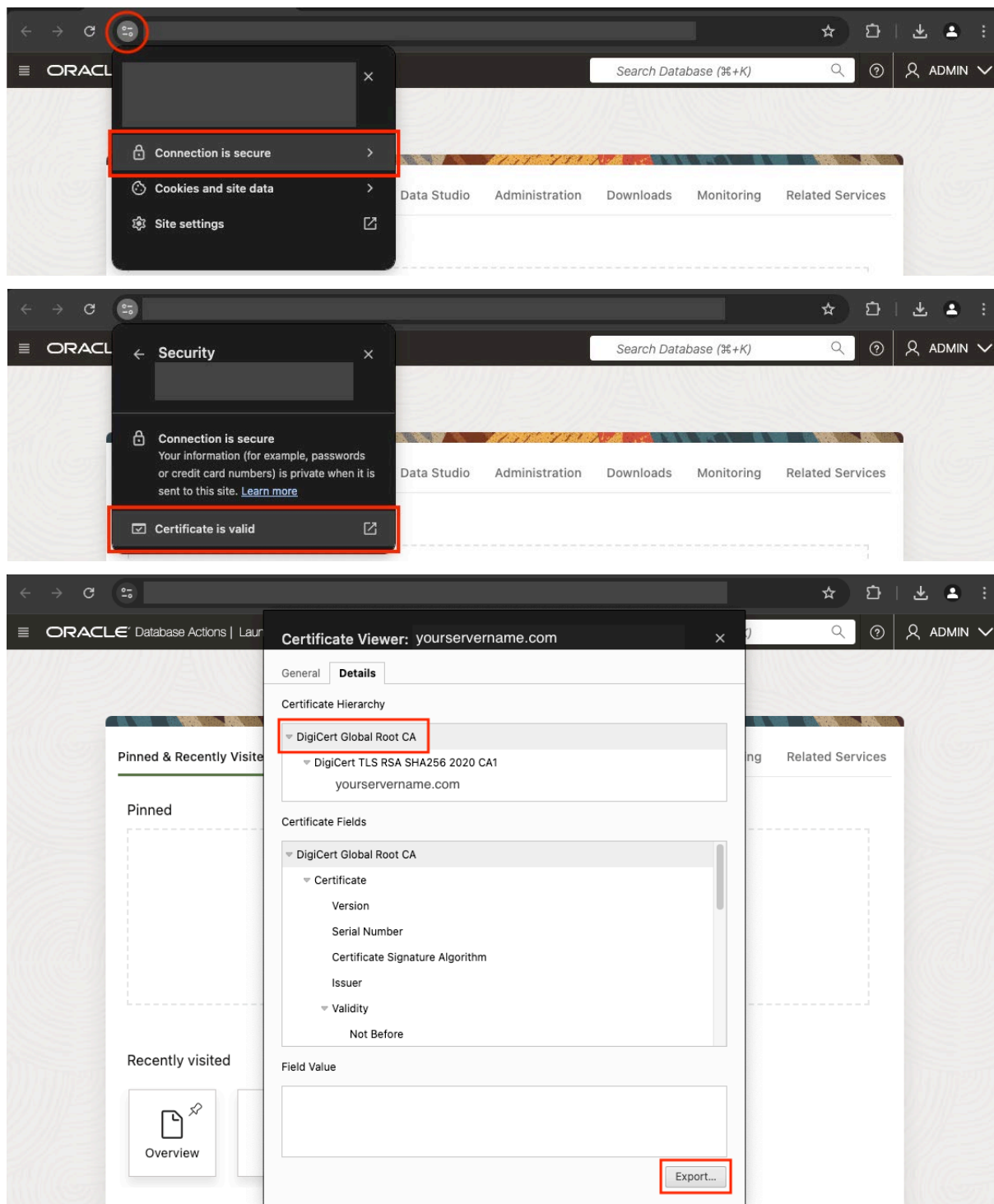
```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE (
    host      => '<autonomous database hostname> ',
    lower_port => 443,
    ace       => xs$ace_type (
      privilege_list => xs$name_list('http'),
      principal_name => apex_application.g_flow_schema_owner,
      principal_type => xs_acl.ptype_db));
END;
```

**Note:** Your Autonomous Database hostname can be the database's fully qualified domain name or its endpoint IP address. The information is available on the **Autonomous Database Details** page in the OCI Console.

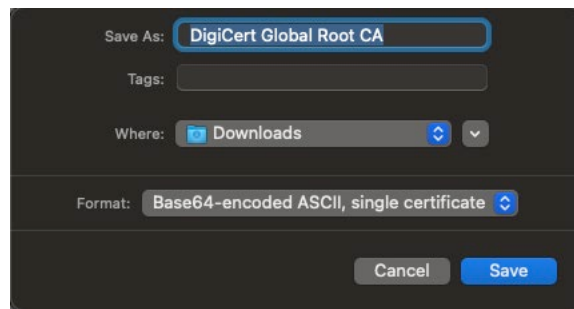
## 5. Configure the Database Wallet.

For your APEX application to invoke the RESTful Web Services deployed in your Autonomous Database, you must add the root CA certificate of the Autonomous Database into the Oracle Wallet in the E-Business Suite database.

To get the root certificate, use your browser to open the Autonomous Database admin console, view the certificate details, and save the root certificate, as shown below using Google Chrome as an example.







After saving the root certificate, connect to your E-Business Suite database server to create the wallet and add the certificate using the **orapki** utility, as shown below.

Source the database environment file.

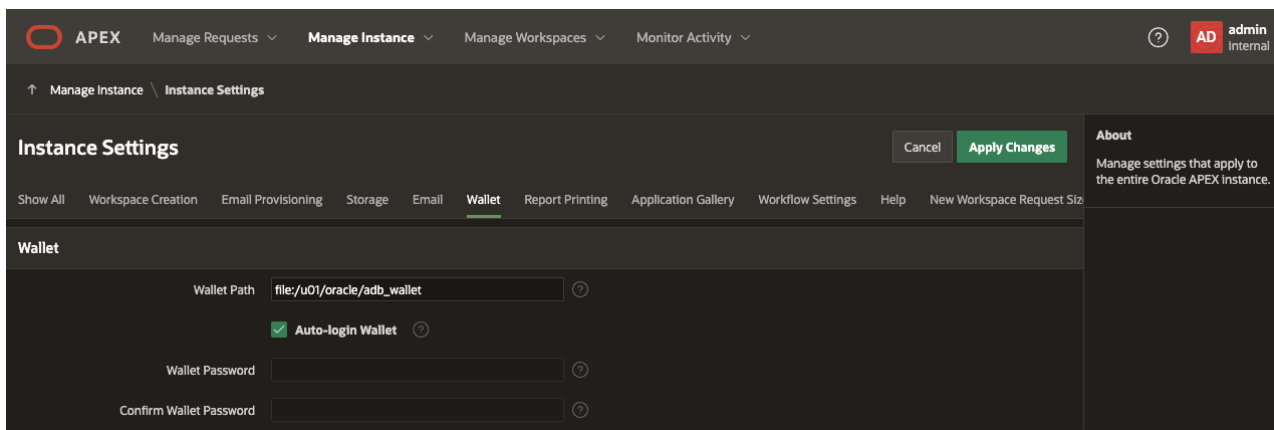
```
orapki wallet create -wallet <wallet file directory> -auto_login_only

orapki wallet add -wallet <wallet file directory> -auto_login_only -trusted_cert -
cert <certificate file>
```

For more information, see [Adding a Trusted Certificate to an Oracle Wallet](#).

## 6. Configure the wallet in the APEX Workspace.

1. From Oracle APEX Administration Services, select **Manage Instance** and then **Instance Settings**.
2. In the **Wallet** tab:
  - Enter the **Wallet Path**, using the following format:  
`file:<directory-path>`
  - Enable **Auto-login Wallet**.
3. Click **Apply Changes**.



## 7. Create an APEX Web Credential for the OAuth Client.

1. Log in to APEX (<https://<EBS hostname>:<port>/ords/apex>) as the administrator.  
For Workspace, enter **EBS\_NLQ\_APEX** and enter the administrator username and password.
2. From **App Builder**, click **Workspace Utilities**.
3. Click **Web Credentials**.
4. Click **Create**.
5. In the following fields, enter:
  - **Name:** EBS NLQ OAuth Credential

- **Static ID:** EBS\_NLQ\_OAUTH\_CREDENTIAL
  - **Authentication Type:** OAuth2 Client Credentials
6. Enter the Client ID and Client Secret created in [Step 8 Create OAuth Client](#) of [Configuring the Oracle Autonomous Database](#).
  7. In the **Valid for URLs** field, enter:  
 https://<autonomous database hostname>/ords/ebsnlq/oauth/token  
 https://<autonomous database hostname>/ords/ebsnlq/ask-orcl
  8. Click **Create**.

The screenshot shows the 'Web Credentials' page in the Oracle APEX interface. The page has a dark theme. At the top, there's a navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. Below this is a breadcrumb trail: 'Workspace Utilities > Web Credentials > Create/Edit'. The main section is titled 'Web Credentials' and has a 'Cancel' button and a green 'Create' button. The 'Attributes' section contains the following fields:

- Name:** EBS\_NLQ OAuth Credential
- Static ID:** EBS\_NLQ\_OAUTH\_CREDENTIAL
- Authentication Type:** OAuth2 Client Credentials (dropdown)
- OAuth Scope:** (empty text field)
- Client ID or Username:** (empty text field)
- Client Secret or Password:** (password field with dots)
- Verify Client Secret or Password:** (password field with dots)
- Valid for URLs:** A large text area containing two lines of text: 'https://<autonomous database hostname>/ords/ebsnlq/oauth/token' and 'https://<autonomous database hostname>/ords/ebsnlq/ask-orcl'. Below this field is a note: 'Note that changing this value requires re-entering the client secret.'

The 'Advanced' section at the bottom has the following fields:

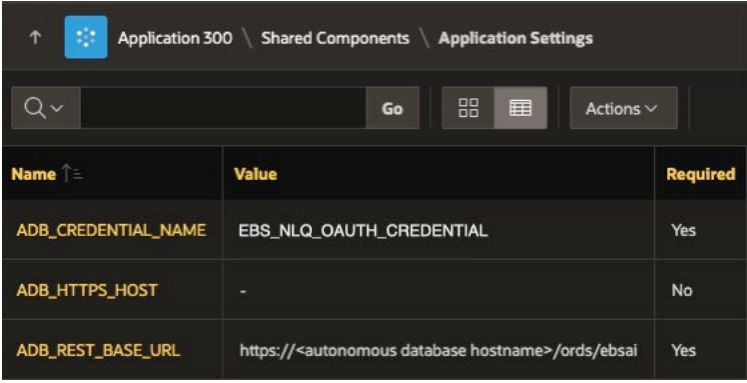
- Prompt On Install:** On (checkbox)
- Comments:** (empty text area)

On the right side, there's a sidebar with the title 'Web Credentials' and the following text: 'Store authentication credentials for external REST services or REST Enabled SQL services. The Client Secret will be stored encrypted, and it can only be used by Oracle APEX and cannot be retrieved in clear text. Also, credential information will not be included in application export files. After importing an application into the target workspace, prompts will be displayed to re-enter the credentials.'

## 8. Update the Oracle E-Business Suite Natural Language Query APEX application settings.

Update the APEX application settings as described in the following steps:

1. Log in to APEX (https://<EBS hostname>:<port>/ords/apex) as the administrator. Enter Workspace **EBS\_NLQ\_APEX** and enter the administrator username and password.
2. From **App Builder**, select the **Oracle E-Business Suite Natural Language Query** application.
3. From the navigation menu, select **Shared Components**, then **Application Settings**.
4. Update the value for ADB\_REST\_BASE\_URL to point to the Autonomous Database ORDS URL.
5. Verify that the value for the ADB\_CREDENTIAL\_NAME is the Static ID of the APEX Web Credential created in the previous [step](#).



| Name                | Value                                             | Required |
|---------------------|---------------------------------------------------|----------|
| ADB_CREDENTIAL_NAME | EBS_NLQ_OAUTH_CREDENTIAL                          | Yes      |
| ADB_HTTPS_HOST      | -                                                 | No       |
| ADB_REST_BASE_URL   | https://<autonomous database hostname>/ords/ebsai | Yes      |

1 The image shows the Application Settings window.

6. (Optional) Enter the value for ADB\_HTTPS\_HOST.

- This represents the host name to be matched against the common name (CN) of the Autonomous Database's certificate for an HTTPS request.
- This value is passed to the p\_https\_host parameter in the apex\_web\_service.make\_rest\_request PL/SQL function call.
- For more information, see [MAKE REST REQUEST Function](#).
- For example, this value is required if you decide to use the IP address for the ADB\_REST\_BASE\_URL setting.

9. (Conditional) Update the Oracle E-Business Suite Natural Language Query APEX application Browser Security attributes.

Completion of this step enables you to embed the APEX application within an E-Business Suite application page. If your APEX server is in a different domain or uses a different port than the E-Business Suite application server, then you must perform this step.

1. Log in to APEX (https://<EBS hostname>:<port>/ords/apex) as the administrator.

For Workspace, enter **EBS\_NLQ\_APEX** and enter the administrator username and password.

2. From **App Builder**, select the **Oracle E-Business Suite Natural Language Query** application.
3. From the navigation menu, select **Shared Components**, then **Security Attributes**.
4. Under **Browser Security**, in the **HTTP Response Headers** field, enter:

```
Content-Security-Policy: frame-ancestors 'self' https://<EBS hostname>:<port>;
```

5. Click **Apply Changes**.

APEX

App Builder

SQL Workshop

Team Development

Gallery

Search

AD

admin

ebs\_ai\_apex

Application 300

Edit Security Attributes

Edit Page 1

Definition

Security

Globalization

User Interface

Progressive Web App

AI

Application 300

Cancel

Apply Changes

Show All

Authentication

Authorization

Session Management

Session State Protection

Browser Security

Database Session

Advanced

Browser Security

Cache

Disabled

Embed in Frames

Deny

Referrer Policy

strict-origin

HTML Escaping Mode

Extended

HTTP Response Headers

Content-Security-Policy: frame-ancestors 'self' https://<EBS\_hostname>:<port>;

About

Use this page to set application-wide security settings. Edit application components directly to manage more granular settings.

## Integrating APEX with E-Business Suite

### 1. Define the Oracle E-Business Suite Profile.

Update the FND: APEX URL profile option with the correct setting at the site level.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **System Administrator**, then **Profile**, and then **System**.
3. Search for the Profile **FND: APEX URL**.
4. Enter Site **https://<EBS hostname>:<port>/ords**.
5. Click **Save**.

The screenshot shows the 'System Profile Values' window. It contains a table with columns: Profile Option Name, Site, Application, Responsibility, and User. The first row shows 'FND: APEX URL' in the Profile Option Name column and 'hx1.oraclevcn.com:8080/ords' in the Site column. The other columns are empty.

| Profile Option Name | Site                        | Application | Responsibility | User |
|---------------------|-----------------------------|-------------|----------------|------|
| FND: APEX URL       | hx1.oraclevcn.com:8080/ords |             |                |      |

### 2. Define an E-Business Suite Form Function.

Define a function that points to the Oracle E-Business Suite Natural Language Query APEX application.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **Functional Administrator**, then **Core Services**, and then **Functions**.
3. Click **Create Function**.
4. Enter the following details:
  - Name: **Ask Oracle E-Business Suite (EBS)**
  - Code: **XX\_NLQ\_APEX\_ASK\_ORCL**
  - Description: **Ask Oracle E-Business Suite (EBS)**
  - Type: **Apex**

The screenshot shows the 'Create Function: Define Function' window. It has a 'Cancel' button and a 'Continue' button. Below the buttons, there is a list of fields with asterisks indicating required fields. The fields are: Name (Ask Oracle E-Business Suite (EBS)), Code (XX\_NLQ\_APEX\_ASK\_ORCL), Description (Ask Oracle E-Business Suite (EBS)), Type (Apex), Maintenance Mode Support (None), and Context Dependence (Responsibility).

**Create Function: Define Function** Cancel Continue

\* Indicates required field

\* Name: Ask Oracle E-Business Suite (EBS)

\* Code: XX\_NLQ\_APEX\_ASK\_ORCL

Description: Ask Oracle E-Business Suite (EBS)

Type: Apex

Maintenance Mode Support: None

Context Dependence: Responsibility

5. Click **Continue**.
6. Select the following details:
  - Layout: **Embedded**
  - Workspace: **EBS\_NLQ\_APEX**
  - Application: **Oracle E-Business Suite Natural Language Query**
  - Page: **Ask Oracle**

**Create Function: Details** Cancel Back Submit

\* Indicates required field

**Apex**

\* Layout Embedded ▾

\* Workspace EBS\_NLQ\_APEX ▾

\* Application Oracle E-Business Suite ▾

\* Page Ask Oracle ▾

| Name              | Value |
|-------------------|-------|
| No results found. |       |

**Web**

\* HTML Call GWY.jsp?targetAppType=APEX&apexLayout=EMBEDDED&p=300:2

☐ Secured

☐ Encrypt Parameters

Host Name

Agent Name

Icon

7. Click **Submit**.

### 3. Define E-Business Suite Menus and Responsibilities.

Attach the function for the Oracle E-Business Suite Natural Language Query APEX application to the appropriate **Menus** and **Responsibilities** within the E-Business Suite environment.

The following example shows how to add the function to an existing menu.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **System Administrator**, then **Application**, and then **Menu**.
3. Search for the desired **Menu** and then add a new entry for the function using the following example field values:
  - **Seq:** 5
  - **Prompt:** Ask Oracle E-Business Suite (EBS)
  - **Function:** Ask Oracle E-Business Suite (EBS)
4. Click **Save**.

**Menus**

Menu MFG\_DIST\_SUPERUSER\_GUI View Tree...

User Menu Name Manufacturing and Distribution Super User

Menu Type Standard

Description Manufacturing and Distribution Super User

Icon

| Seq | Prompt           | Submenu      | Function                | Description | Grant                               |
|-----|------------------|--------------|-------------------------|-------------|-------------------------------------|
| 10  | Inventory        | INV_NAVIGATE |                         |             | <input checked="" type="checkbox"/> |
| 5   | Ask Oracle E-Bus |              | Ask Oracle E-Business S |             | <input checked="" type="checkbox"/> |

You can now launch the Oracle E-Business Suite Natural Language Query APEX application from the appropriate Menu and Responsibility as shown in the following screenshot.

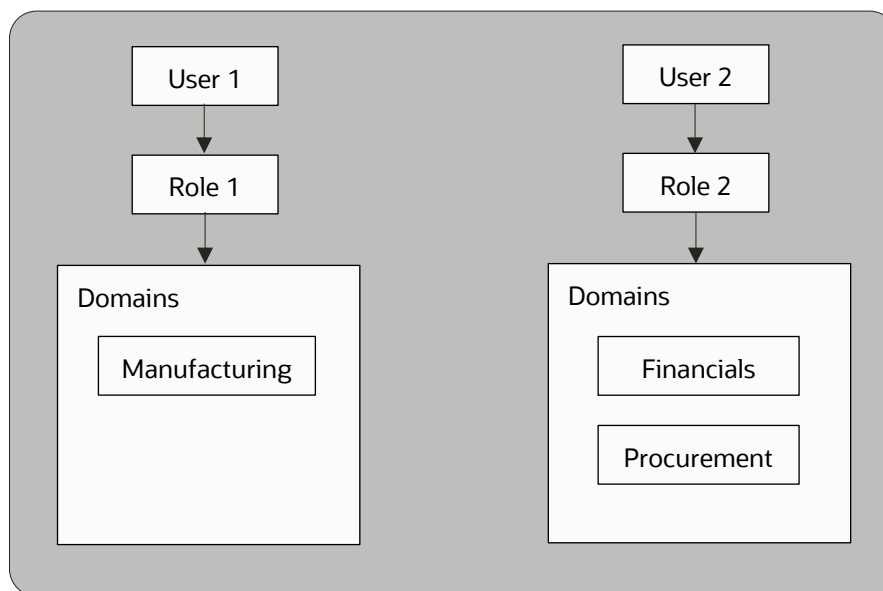
## Home

- Manufacturing and Distribution Manager
- Ask Oracle E-Business Suite (EBS)

## Configuring Security

Each question is executed against a domain, and each domain has a corresponding AI Profile. You can assign different roles to different users. Each role provides a user with access to a specific domain.

For example, you can create a role with access only to the Manufacturing domains and another role with access to Financials and Procurement domains. Assign each role to a different user, as shown in the following graphic.



### 1. Verify Permissions.

Verify that the permissions created by the installation scripts are visible in Oracle E-Business Suite.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **Functional Administrator**, then **Security**, and then **Permissions**.
3. Search for Code **EBS\_NLQ%**.



ORACLE Applications Administration
 

[Home](#)
[Favorites](#)
[Settings](#)
[Worklist](#)
Logged In As SYSADMIN
[Help](#)
[Logout](#)

---

Security
[Core Services](#)
[Personalization](#)
[File Manager](#)
[Portletization](#)
[Configuration Manager](#)
[Allowed Resources](#)
[Allowed Forwards](#)

Grants
Permissions
[Permission Sets](#)

## Permissions Save Search

### Simple Search

Note that the search is case insensitive Advanced Search

Name

Code

Object Name 🔍

| Name ▲                                               | Code ▲                 | Object Name ▲ | Last Update ▲ | Duplicate | Update |
|------------------------------------------------------|------------------------|---------------|---------------|-----------|--------|
| EBS NLQ permission for Discrete Costing domain       | EBS_NLQ_DMFCST_PROFILE |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Discrete Manufacturing domain | EBS_NLQ_DMF_PROFILE    |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Dispute domain                | EBS_NLQ_AR_PROFILE     |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Logistics domain              | EBS_NLQ_INV_PROFILE    |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Maintenance domain            | EBS_NLQ_EAM_PROFILE    |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Order Management domain       | EBS_NLQ_ONT_PROFILE    |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Payment domain                | EBS_NLQ_AP_PROFILE     |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for People domain                 | EBS_NLQ_HCM_PROFILE    |               | 10-Oct-2024   |           |        |
| EBS NLQ permission for Process Manufacturing domain  | EBS_NLQ_GME_PROFILE    |               | 10-Oct-2024   |           |        |

Create Permission

...
Rows 1 to 13

## Create Permission Set

Cancel Apply

\* Indicates required field

|             |                                                          |
|-------------|----------------------------------------------------------|
| * Name      | EBS NLQ permission set for Discrete Manufacturing domain |
| * Code      | XX_NLQ_DMF_PS                                            |
| Description | EBS NLQ permission set for Discrete Manufacturing domain |

## Permission Set Builder

| Select Rows:             | Remove                            | + ...       |
|--------------------------|-----------------------------------|-------------|
| Permission Set           | Permission                        | Description |
| <input type="checkbox"/> | EBS NLQ permission for Discrete M |             |

### 3. Define roles and grants.

Create roles and grants according to your functional and security requirements. For example, you can create a role with access to a single domain, or you can create a role with access to multiple domains.

The following example demonstrates how to create a role and grant for the Discrete Manufacturing domain.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **User Management**, then **Roles & Role Inheritance**.
3. Click **Create Role**.
4. Enter the following field values:
  - **Role Code:** UMX|XX\_NLQ\_DMF\_ROLE
  - **Display Name:** EBS NLQ role for Discrete Manufacturing domain
  - **Description:** EBS NLQ role for Discrete Manufacturing domain
  - **Application:** Manufacturing

## Create Role

Cancel Security Wizards Save Apply

\* Indicates required field

|                |                                              |               |               |
|----------------|----------------------------------------------|---------------|---------------|
| * Category     | Miscellaneous                                | * Application | Manufacturing |
| * Role Code    | UMX XX_NLQ_DMF_ROLE                          | * Active From | 03-Sep-2024   |
| * Display Name | EBS NLQ role for Discrete Manufacturing doma | * Active To   |               |
| * Description  | EBS NLQ role for Discrete Manufacturing doma |               |               |

## Permissions

| Create Grant      | ... |        |                   |               |             |           |        |        |
|-------------------|-----|--------|-------------------|---------------|-------------|-----------|--------|--------|
| Name              | Set | Object | Data Context Type | Access Policy | Last Update | Duplicate | Update | Delete |
| No results found. |     |        |                   |               |             |           |        |        |

5. Click **Create Grant**.
6. Click **Save and Proceed**.
7. Enter the following field values:
  - **Name:** EBS NLQ grant for Discrete Manufacturing domain
  - **Description:** EBS NLQ grant for Discrete Manufacturing domain
8. If desired, enter a **Responsibility** under the **Security Context** region.

## Create Grant: Define Grant

Cancel Step 1 of 4 Next

\* Indicates required field

|                  |                                                 |
|------------------|-------------------------------------------------|
| * Name           | EBS NLQ grant for Discrete Manufacturing domain |
| Description      | EBS NLQ grant for Discrete Manufacturing domain |
| * Effective From | 03-Sep-2024                                     |

Effective To

## Security Context

Define the context when the grant is applied by selecting a grantee, a responsibility and/or operating unit.

|                |                                                |
|----------------|------------------------------------------------|
| Grantee Type   | Group Of Users                                 |
| Grantee        | EBS NLQ role for Discrete Manufacturing domain |
| Operating Unit |                                                |
| Responsibility |                                                |

## Data Security

To define a data security policy, select an object.

|        |  |
|--------|--|
| Object |  |
|--------|--|

9. Click **Next**.

10. In the **Set** field, enter **EBS NLQ permission set for Discrete Manufacturing domain**.

## Create Grant: Define Object Parameters and Select Set

Cancel Back Step 3 of 4 Next

\* Indicates required field

## Set

Select the permission set or menu navigation set that defines the grantee's access.

|       |                                                          |
|-------|----------------------------------------------------------|
| * Set | EBS NLQ permission set for Discrete Manufacturing domain |
|-------|----------------------------------------------------------|

11. Click **Next**.

12. Click **Finish**.

## 4. Assign Roles.

Assign one or more EBS NLQ roles to the user.

The following example demonstrates how to assign the Role **EBS NLQ role for Discrete Manufacturing domain** to a user.

1. Log in to E-Business Suite as the SYSADMIN user.
2. From the **Home** page, click **User Management**, then **Users**.
3. Search for a user, and then click **Update**.
4. Click **Assign Roles**.
5. Search for **EBS NLQ role for Discrete Manufacturing domain** and select the **Role**.
6. Enter a **Justification**.
7. Click **Apply**.

## Update User: mfg

Cancel Reset Password Save Apply

\* Indicates required field

Prefix  
First Name Jonathan  
Middle Name  
Last Name Smith  
Suffix

\* User Name mfg  
Email nobody@localhost  
Status  Active  
\* Active From 01-Jan-1996   
Active To 

### Quick Tips

Personal information originates from the HR system and cannot be updated here.

Roles Contact Information Securing Attributes

Changes can only be made for roles for which you have been granted administrative privileges.

Assign Roles

Search
All Roles
GO

Rows 1 to 30

| Details                                                                           | Role  | Description                                    | Status  | Remove                                                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | EBS NLQ role for Discrete Manufacturing domain                                         | EBS NLQ role for Discrete Manufacturing domain | Ready for Submission                                                                      |  |

\* Justification
Assign EBS NLQ role

\* Active From
15-Aug-2024

Active To

Role Inheritance

## Performing Oracle E-Business Suite Administrative Steps

### 1. Verify or set the Costing Domain profile option.

The profile option **CST: Endeca Txn Reg Start Date (DD-MON-YYYY)** enables the administrator to set a cutoff date for the transaction accounting they wish to query. The system retrieves accounting data from this cutoff date onward. For optimal performance, set the date to cover a time horizon of one quarter (3 months). If no value is specified for the profile, the default cutoff is set to 100 days.

### 2. Set up the HRMS security profile for the People domain.

To get query results for the People domain, perform the following one-time setup steps:

1. Set the profile option **HR: Extension For Endeca or ECC** to **Yes**.
2. Following the [Reviewing User's Security Profiles](#) section in the *Oracle HRMS Implementation Guide*, use the **HRMS Security Profile** window to create a security profile, or select an existing one, and set the extension value to **Endeca or ECC**.
3. Use the **HRMS Assign Security Profile** window to assign the security profile to a User or Responsibility according to your business policy.
4. Run the *Security List Maintenance* concurrent program after assigning the security profile.
5. Run the *Populate Shared Tables for Person Search* concurrent program. Select the extension value as **Endeca or ECC**.
6. (Recommended) Schedule the concurrent program in Step 5 to run in regular intervals.

### 3. Schedule the Quality data load concurrent program.

If you have already configured the [Quality Command Center](#) and scheduled the [Quality Command Center Data Load](#) concurrent program to run periodically using the **Incremental data load** mode, then you may skip this step.

Otherwise, perform the following steps:

1. Submit the *Publish Quality Collection Metadata to Enterprise Command Center* concurrent program. See [Publishing Quality Collection Metadata to Enterprise Command Center](#) in the *Oracle Quality User's Guide*.
2. Schedule the concurrent program in Step 1 to run at regular intervals.
3. (Conditional) If quality plans have been created or updated, submit the *Publish Quality Collection Metadata to Enterprise Command Center* concurrent program again.

## Troubleshooting

### Test with Select AI's Chat Mode

If the application does not produce any results, perform a standalone test using the standard *Select AI* chat mode. This verifies the OCI Resource Principal authentication configuration has been correctly configured.

Connect to the Autonomous Database as the **XX\_NLQ\_UTIL** user. Run the following SQL statements:

```
-- Create a test AI Profile
BEGIN
  DBMS_CLOUD_AI.CREATE_PROFILE (
    profile_name => 'TESTGENAI',
    attributes => '{"provider": "oci",
                  "credential_name": "OCI$RESOURCE_PRINCIPAL",
                  "object_list": [{"owner": "XX_NLQ_UTIL", "name": "DUAL"}],
                  "model": "meta.llama-3.1-70b-instruct",
                  "temperature": "0.4",
                  "oci_apiformat": "GENERIC"
                  }'
  );
END;
/
```

Since an LLM model may be deprecated and replaced by a more advanced version, refer to the *Chat Models* section in the [Pretrained Foundational Models in Generative AI](#) OCI documentation topic. If the `meta.llama-3.1-70b-instruct` model is no longer available, replace the “model” value above with the current version.

```
-- Try the standard Select AI chat mode
SELECT DBMS_CLOUD_AI.GENERATE(prompt      => 'What is Oracle DB',
                              profile_name => 'TESTGENAI',
                              action      => 'chat')

FROM DUAL;
```

If you encounter an ORA-20404: Object not found error, it is possible the resource principal or dynamic group has not been activated yet. Due to caching, it may take up to two hours for your changes to take effect.

### See also

- [Configure OCI Generative AI](#)
- [Perform Prerequisites to Use Resource Principal with Autonomous Database](#)
- [Integrating OCI Generative AI with Select AI and APEX to query data using natural language](#)

### Check the Dynamic Group Matching Rule Syntax

The correct syntax to create a dynamic group matching rule for your Autonomous Database is to use `resource.id`.

Correct:

```
resource.id = '<your-autonomous-database-ocid>'
```

Incorrect:

```
instance.id = '<your-autonomous-database-ocid>'
```

## Diagnose Network Connection from EBS DB to Autonomous DB

If the standalone chat mode test case is successful but you are not getting results in APEX, diagnose the network connection by executing the following query in the EBS DB's APPS schema:

```
SET serveroutput ON

DECLARE
    l_response_clob CLOB;
BEGIN
    l_response_clob := apex_web_service.make_rest_request(
        p_url           => '<ADB_REST_BASE_URL>',
        p_http_method   => 'GET',
        p_https_host     => '<ADB_HTTPS_HOST>'
    );
    DBMS_OUTPUT.PUT_LINE('response :'||l_response_clob);
END;
/
```

Replace the variables `<ADB_REST_BASE_URL>` and `<ADB_HTTPS_HOST>` with values you have provided in the previous step [Update the Oracle E-Business Suite Natural Language Query APEX application settings](#).

A successful response may look like this:

```
response :{
  "code": "NotAuthorizedOrNotFound",
  "title": "Procedure not authorized or not found",
  "message": "The procedure named ebsnlq could not be accessed or found",
  "o:errorCode": "ORDS-22001",
  ... shortened for brevity ...
}
```

**Note:** Even though the response contains some error codes, this is considered a successful network connection test.

If the test is not successful, an ORA- error message appears. The following table shows some common network connection issues and their solutions.

| ERROR MESSAGE                                                                 | SOLUTION                                                                                                                                                                                            |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA-24247: network access denied by access control list (ACL)                 | Verify the <code>DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE</code> command described in this <a href="#">step</a> is properly executed.                                                                 |
| ORA-28759: failure to open file                                               | Verify the wallet file path given in this <a href="#">step</a> is spelled correctly and the wallet file is readable.                                                                                |
| ORA-24263: Certificate of the remote server does not match the target address | The digital certificate was not downloaded or imported correctly in this <a href="#">step</a> , or the <code>ADB_HTTPS_HOST</code> variable was not set up correctly in this <a href="#">step</a> . |

## Conclusion

Oracle E-Business Suite provides many powerful querying and reporting features. But most of these querying features are very sophisticated and require in-depth user training to use them effectively. This paper presents a natural language query system for Oracle E-Business Suite. The system leverages Oracle Autonomous Database's Select AI feature and OCI Generative AI which translates an English question into a SQL query. The SQL query is then executed in the EBS database to provide the results. Each query is executed in a new XX\_NLQ schema that has minimum grants. Together with new secured views and new VPD policies, the Oracle E-Business Suite security setup is honored. The system utilizes Oracle APEX to provide a simple and intuitive *Ask Oracle E-Business Suite* user interface.

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